

A Life in Songs: Visualizing Long-Term Spotify Listening History for Personal Reflection

02808 Personal Data Interaction — Group 7

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Abstract

Music streaming platforms collect rich data about users' listening behavior, but offer little support for reflecting on it beyond annual summaries. We present *A Life in Songs*, an Observable Notebook prototype that turns extended Spotify listening history into interactive, temporally navigable visualizations. A seven-participant Think-Aloud usability study led to simplified chart descriptions, reordered sections, clearer legends, and the merging of the genre stream graph into the radial chart. We then ran a within-subjects HCI experiment ($N = 9$ of a planned 12, AB/BA counterbalanced) comparing the visualization alone (V) against the visualization augmented with reflective guidance (VR), on a late-night listening comparison task. Contrary to our hypothesis, the participants were, on average, slower, less accurate, and less confident under VR; the experimental procedure, scenarios, and analysis plan are reported in full. The pattern tentatively suggests that reflective guidance may serve better as one-off onboarding than as a persistent overlay.

Keywords

Personal informatics, music streaming, Spotify listening history, interactive visualization, data exploration, music analytics, Think-Aloud test, HCI experimentation

1 Introduction

1.1 Analysis

The core problem addressed in this project is the limited availability of accessible and meaningful tools for personal reflection on music listening behavior. Although streaming platforms collect vast amounts of listening data, they rarely give users the ability to explore that data freely and on their own terms.

Music listening behavior is temporal; tastes shift between seasons, life events, and moods. A useful tool in this field must support exploration across time, allowing users to identify trends, revisit past periods, and draw their own conclusions. Additionally, the prototype must handle personal data in a way that feels relevant and engaging rather than purely statistical and analytical.

The design challenge, therefore, is to develop an interactive visualization system that is expressive enough to reveal meaningful patterns in listening history, while remaining intuitive and accessible to users without technical expertise.

1.2 Prototype

Through iterative usability testing and successive rounds of re-design, we developed an interactive data-exploration notebook in

Observable that enables users to revisit and reflect on their long-term Spotify listening history. The notebook presents eight coordinated visualizations, each targeting a distinct question about the user's musical identity and how it has changed over time.

1.2.1 Data source and processing. Spotify offers users an *Extended Streaming History* export containing every play event recorded by the platform. Each record includes the track and artist name, a millisecond-precision play duration, a UTC timestamp, the connection country, the platform used, start and end reasons (e.g. clickrow, trackdone), and flags for shuffle, skip, and incognito mode. A single user's export can span more than a decade and contain hundreds of thousands of entries. The curated dataset for this project covers the period from December 2011 to March 2026.

Because the raw export is large and noisy, a three-stage Python pipeline transforms it into visualization-ready datasets. The pipeline is shown in Figure 1.

In the first stage, the pipeline filters out podcasts and audiobooks, drops plays shorter than 30 seconds (Spotify's own threshold for counting a royalty-generating stream) or longer than 60 minutes (a heuristic for stuck sessions), removes incognito plays, and parses timestamps into calendar features such as year-month, day-of-week, and hour.

In the second stage, genre information—a data point absent from the export—is retrieved via the Spotify Web API: unique artists are batch-queried to obtain their genre tags, and each stream is labeled with the primary genre of its artist. This artist-level genre assignment is a deliberate simplification: although an artist's theme and style can vary substantially across albums, artist genres are more consistently available through the Spotify API and provide a stable and comparable classification across the full listening history. Using album-specific genres could, in principle, offer finer granularity, but it would also introduce greater sparsity, inconsistency, and additional API complexity.

In the third stage, the enriched dataset is aggregated into ten compact CSV files, one per visualization family, which the Observable Notebook loads directly. These include genre, artist, daily-intensity, geographic, and behavioral summaries, alongside a set of hour-resolved aggregations that support the temporal visualizations.

1.2.2 Visualizations. The notebook is organized into thematic dashboards, a structure that emerged from usability feedback (see Results). Each visualization is designed around a user question that arose during exploration of the data:

How much did I listen, and when? A GitHub-contributions-style calendar heatmap displays daily listening hours across all years. An interactive year-range selector lets users zoom into specific periods.

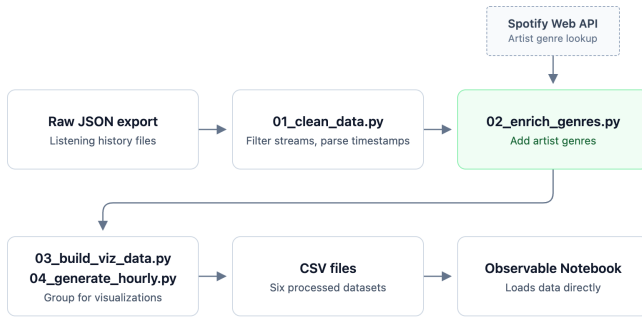


Figure 1: Overview of the data processing pipeline

The 2018–2020 gap (caused by a platform switch) is immediately visible and serves as a starter about life changes.

How has my taste evolved? A radial genre chart lets users select any year and month to see the proportional breakdown of their listening by genre. The shape of the chart changes as genres rise and fall, offering an at-a-glance summary of the user’s sonic identity at any point in time.

Who dominated my listening, and when did that change? A cumulative bar-chart race animates through months, showing how artists (or genres, via a toggle) accumulate listening hours over time. Bars grow, overtake one another, and new entries appear. A separate artist bump chart ranks the top artists per month, with hover-to-highlight and a drop-down filter for isolating individual trajectories.

Am I exploring or replaying? Two discovery-oriented views address listening openness. One tracks the monthly ratio of newly encountered artists to familiar ones; the other breaks down engagement style—active selection versus passive autoplay—alongside shuffle usage over time.

Where was I when I listened? A geographic dashboard maps listening activity by country, highlighting periods of travel outside the user’s home country. Summary cards show total hours abroad, number of countries, and the top destination.

What time of day do I listen? A streaming-hours explorer aggregates plays into a polar clock, distinguishing daytime from late-night listening and allowing period comparisons. This chart also serves as the focal visualization for the HCI experiment described in Section 2.2.4. Three alternatives were sketched and rejected; the rationale is reported in Appendix B.

1.2.3 Design rationale. The design follows the personal informatics framing proposed by Li et al. [5]: Spotify’s platform handles data *collection* passively over years; the Python data pipeline performs *integration* by cleaning, enriching, and reshaping the raw export; and the interactive notebook supports *reflection* by letting users navigate temporal patterns, filter by dimension, and annotate what they find. The system deliberately avoids prescriptive scores or recommendations. The goal is to surface patterns and let users draw their own conclusions, which aligns with the model’s emphasis on user-driven reflection rather than system-driven action.

1.3 Related Work

Several existing tools and projects address aspects of music data visualization and listening analytics, although none fully meet the goals of the present project.

Spotify Wrapped is perhaps the most widely recognized example of personalized music analytics. It is released annually, it presents users with a curated summary of their year in music, including top artists and songs [8]. But, it is static, available only once a year, and offers no ability for users to explore their data freely or across different time periods.

Last.fm takes a different approach, continuously tracking listening activity across platforms and providing detailed statistics on listening history [4]. However, its interface is largely data-driven rather than experiential, and it lacks the kind of narrative or temporal exploration that this project aims to provide.

Last.History, from Baur et al. (2010), is a tool to bridge the gap between Last.fm raw data and the user’s need for a narrative history. It allows the kind of free exploration that is missing from the previous platforms [1].

Silent Data, Wirfs-Brock et al. [9] coin the term *silent data* for information that drives a listening experience but is never surfaced to listeners. Their CHI 2020 workshops imagined voice-assistant interactions over Spotify data; we target the same gap through interactive visualization instead.

Lived Informatics, Epstein et al. [2] extend Li et al.’s stage-based model with a *lived informatics* perspective, accounting for how self-tracking integrates into everyday life rather than narrow behavior-change goals. Our prototype follows this framing: it offers tools to revisit listening as part of lived experience rather than prescribing changes.

2 Method

We used two complementary evaluation methods: a usability evaluation and a human-computer interaction experiment. The usability evaluation examined whether users could understand, navigate, and interpret the prototype, while the experiment tested whether reflective guidance improved performance on a late-night listening task compared with visualization alone. Together, the methods address both interaction quality and measurable task performance.

2.1 Usability Evaluation

The goal of the usability evaluation was to assess whether users could independently read, navigate and extract meaningful information from the interactive visualizations.

To address these questions, we used the concurrent Think-Aloud method [3, 6], in which participants verbalize their thoughts continuously while interacting with the prototype. This approach was chosen because it captures interpretation in real time, making it easier to pinpoint exactly where confusion arises. Since the main visualization can be read differently across users, concurrent verbalization was particularly well suited to identifying misunderstandings and moments of hesitation early in the design process. In this process, we managed to execute this usability evaluation with 7 participants.

One deviation from the standard think-aloud practice is worth noting. In a typical controlled usability study, the evaluator remains

strictly neutral and does not intervene during task completion [6]. In our sessions, however, if a participant had difficulty understanding something or was unable to proceed, we provided minimal guidance to keep the session moving and to gather observations from later sections of the prototype. This trade-off reduced some ecological validity but was necessary given the exploratory and iterative nature of the evaluation.

Participants were selected to reflect a range of familiarity with data visualization and the Observable Notebook environment. Each session lasted approximately 30 minutes and followed the structure presented in Table 1.

Table 1: Think-Aloud Session overview

Time	Phase	What happens
~5 min	Introduction	Briefly explain the idea behind the prototype and the think-aloud method
~10 min	Prototype Run-Through	Participant explores the notebook with guidance from start to finish, thinking aloud
~10 min	Specific Tasks	A small set of quick tasks, one for each chart, to find possible barriers on the prototype
~5 min	User Experience	Short debrief. Open space for the participant to raise anything that was not captured during the session

The tasks were distributed equally in sections and were the same in each session to keep the interaction process consistent. This allowed comparison of performance across different visualizations equally. The tasks are included in Appendix C. Repeated delays or failures across participants, regarding the same tasks were treated as indicators that a given visualization or its accompanying explanation required revision. Throughout the development process, multiple evaluation sessions were conducted to iteratively refine the prototype, with each round of changes shaping the next [7].

2.2 HCI Experiment

While the usability evaluation focused on whether users could read and navigate the prototype, the HCI experiment was designed to test a performance-oriented question. Following Hornbæk et al.’s [3, Ch. 43] and MacKenzie’s guidance [6], we framed our research question as: *When users interpret late-night patterns in a Spotify listening history, does the addition of reflective guidance to the visualization reduce error rate, decrease decision time, and increase user confidence, compared to the visualization alone?* We hypothesized that adding textual reflective prompts would reduce decision time, lower the error rate, and increase confidence.

2.2.1 Participants. We planned to recruit 12 regular Spotify users, divided evenly between the two counterbalancing groups (six per group), consistent with sample sizes reported in similar within-subjects HCI experiments. Inclusion criteria are: at least one year of continuous Spotify use, willingness to take part in a short observational study, and no prior exposure to the prototype. Recruitment is by convenience sampling through DTU networks. Pre-questionnaire data (age, daily Spotify usage, prior use of Wrapped

or Last.fm, and self-rated comfort with charts) are collected to describe the sample. The sample is summarized in Appendix D.4.

2.2.2 Apparatus. Sessions run on a single laptop with a fixed window size, so hardware and rendering are constant across participants. The two conditions differ only in a single **focal chart** of the Observable Notebook:

- **Visualization-only (V):** the focal chart is shown without textual reflective prompts.
- **Visualization and reflective guidance (VR):** the same focal chart is augmented with short reflective prompts and contextual annotations.

All participants see the same shared group dataset—the aggregated listening history of the project group—which removes the between-subjects noise that personal Spotify exports would introduce.

2.2.3 Procedure. Each session lasts approximately 30 minutes and follows a fixed sequence (full timing breakdown in Appendix 2). After completing a short questionnaire, the participant receives a verbal explanation of the task and then completes a test under the condition that comes first. They then complete three scenarios under their first interface (V or VR, depending on group assignment), followed by three further scenarios under the second interface using the parallel scenario set. Decision time and answers are logged automatically, and confidence is collected on a 5-point Likert scale immediately after each answer. The session closes with a short post-questionnaire on perceived helpfulness of each interface and an open debrief.

The core task is: *given two specified time periods shown side by side (e.g., 2021–2022 vs. 2023–2024), decide which contains more late-night listening activity.* Late-night is categorized as listening between **22:00 and 04:00**, in line with the band marked on the polar visualization. Timing begins when the scenario is shown and ends when the participant submits an answer. An error is defined as any submitted answer that does not match the ground truth derived directly from the dataset.

2.2.4 Design. The experiment uses a **within-subjects design** with one independent variable, **interface type**, at two levels: visualization-only (V) and visualization with reflective guidance (VR). Three dependent variables are measured: **decision time** (seconds), **error rate** (binary per trial, aggregated per condition), and **confidence** (5-point Likert). Within-subjects was chosen for the standard HCI reasons [6, Ch. 5]; the order is counterbalanced via the simplest 2-level Latin square, with half the participants seeing V then VR (group AB) and half seeing VR then V (group BA).

Two risks were anticipated before testing. First, because the task is **knowledge-based**, a participant who has compared two periods cannot meaningfully re-answer the same comparison; we therefore prepared two parallel scenario sets of comparable structure and difficulty, one per condition. Second, **asymmetric skill transfer** may occur because VR is a richer version of V: the reflective prompts may teach a way of reading the chart that carries into V on the second half of testing, while V → VR offers no comparable transfer. We check this at analysis time by comparing the AB and BA group means; if they diverge noticeably, the V vs. VR comparison must be interpreted with caution. Beyond these two risks, several *nuisance*

factors [3]/confounding variables [6] — prior experience with Spotify analytics, visualization literacy, and prior exposure to reflective prompts — are captured in pre-questionnaire.

Excluding practice, the planned design yields 12 participants $\times$ 2 conditions $\times$ 3 scenarios = 72 trials; the realized $N = 9$ sample is reported in Section 3.2. The full analysis plan is given in Appendix D.3.

3 Results and Discussion

3.1 Usability Evaluation: Think-Aloud Method

The early sessions prompted significant changes to the prototype. The first concerned the written content because most users struggled with the technical descriptions accompanying each chart, often pausing for extended periods of time or skipping them entirely. This indicated that the explanations were not effectively supporting interpretation. All descriptions were consequently rewritten in simpler, more direct language and re-framed to highlight what the user should look for rather than how the chart was constructed. The overall writing style was also shifted toward a more narrative tone, encouraging exploration and maintaining engagement throughout.

The second wave addressed the visualizations themselves. Several participants expressed confusion not only about how to read certain charts but also about what they were meant to take away from them. For example, in the first version of the prototype we had the graph in Figure 2, but users found it quite complicated to understand and overcrowded, so we changed it for a simpler and clearer visualization (in Figure 3). The most cognitively demanding visualizations were simplified, redundant elements removed and short guiding captions added to make the purpose of each chart explicit. Interaction cues such as hover hints and filter instructions were made more visible. Beyond individual charts, a recurring concern was the sheer volume of content. Many participants felt overwhelmed by the number of visualizations presented in sequence. In response, related charts were grouped into thematic dashboards, compartmentalizing the material by topic and making the notebook considerably more structured and dynamic.

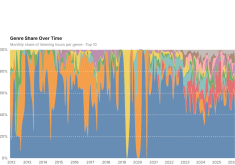


Figure 2: Stream graph before

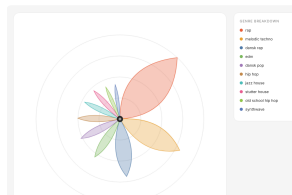


Figure 3: Stream graph after

One methodological limitation of the evaluation concerned the task design. Although tasks were written to be as clear as possible, several participants found the wording ambiguous and asked for clarification mid-session, which reduced the validity of some observations since guided completions are harder to interpret than independent ones. Session timing also proved difficult to control. The free-exploration phases sometimes ran longer than planned, compressing the time available for structured tasks. Both issues

were partially addressed in later sessions by simplifying task wording and introducing soft time prompts during the run-through phase.

Throughout the development process, multiple usability evaluations were conducted to iteratively refine the prototype. The more detailed findings are summarized in Appendix C for each participant, including their profile, observed behavior and the resulting design changes (with some images of the old versions of our prototype).

3.2 HCI Experiment

We report preliminary results from $N = 9$ participants (4 in order $V \rightarrow VR$, 5 in order $VR \rightarrow V$); time constraints prevented us from reaching the planned $N = 12$. On accuracy, five of nine participants tied across conditions, leaving only four informative pairs and a minimum two-sided p -value of 0.125.

Decision time. Mean decision time was 4.10 s ($SD = 3.31$) under V and 6.66 s ($SD = 6.26$) under VR — VR was about 62% slower. Differences were not normally distributed (Shapiro-Wilk $W = 0.66$, $p < 0.001$). One participant's VR session ran past 20 s and drove most of the non-normality. Because the distributional assumptions required for a paired t -test were therefore violated [3], we used the Wilcoxon signed-rank as planned [6], which returned $W = 7$, $p = 0.074$. Standard deviations are large relative to the condition gap, and per-participant ranges overlap substantially.

Error rate. The trial-level error rate was 7.4% under V and 14.8% under VR. At the per-participant level, mean accuracy was 0.926 under V and 0.852 under VR, but five of nine participants had identical accuracy in both conditions; the Wilcoxon returned $W = 2.5$, $p = 0.625$. Three participants were more accurate under V, one was more accurate under VR.

Confidence. Mean self-reported confidence 4.52/5 under V and 4.22/5 under VR (Wilcoxon $W = 4.5$, $p = 0.141$). Of the three measures, decision time shows the strongest direction and the smallest p -value; accuracy and confidence point the same way but with smaller per-participant gaps.

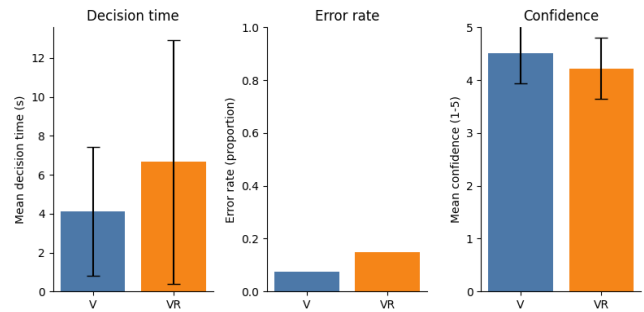


Figure 4: Per-participant means with ± 1 SD whiskers for the three dependent variables under V and VR ($N = 9$). VR was on average slower, less accurate, and rated less confidently than V; none of the three differences reached $p < 0.05$.

Counterbalancing check. The AB group ($n=4$) produced near-identical means across conditions ($V = 5.46$ s, $VR = 5.57$ s). The BA group ($n = 5$) was markedly faster under V (3.02 s) than under

VR (7.54 s), which they encountered *first*. This pattern matches the asymmetric skill transfer pre-registered as a risk in Section 2.2.4: BA participants appear to use VR as a one-off learning aid that then accelerates performance on V, but this same advantage does not transfer in the opposite direction.

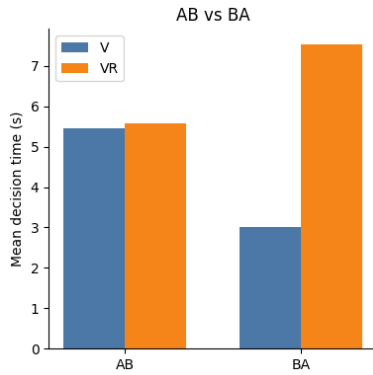


Figure 5: Mean decision time by counterbalancing group. AB participants (V then VR) show nearly equal times across conditions; BA participants (VR then V) are markedly faster under V, consistent with asymmetric skill transfer where exposure to VR sped up later V performance, but the reverse is not observed.

Discussion. All three measures point against the hypothesis: under VR, participants were, on average, slower, less accurate, and less confident. Three caveats temper this reading. First, $N=9$ is small: the Wilcoxon on decision time falls just short of conventional significance ($p = 0.074$), and the accuracy test has insufficient power once ties are accounted for. Second, the asymmetric skill transfer above means part of V’s apparent advantage is an order artifact, not a clean property of the interface. Restricted to the AB group, V and VR are essentially equal in decision time. Third, scenario B1 had only 67% accuracy (6/9) against 78–100% for the other six scenarios; as detailed in Appendix D.5, B1 is the only scenario in which Period 1 is chronologically later than Period 2, suggesting it measured adherence to a “left = earlier” expectation as much as late-night identification. Because B1 sits in Set B and Set B is always seen under VR, B1’s errors fall entirely in the VR column.

B1 sensitivity check. To quantify B1’s impact on the result, we recomputed the per-condition means with B1 excluded (the VR side then averages B2 and B3 only). With B1 removed, the accuracy gap *reverses*: $V = 0.926$, $VR = 0.944$. The confidence gap shrinks from +0.30 to +0.07 Likert points ($V = 4.52$, $VR = 4.44$). The decision time gap also shrinks but stays in V’s favor, going from -2.56 s to -1.54 s ($V = 4.10$ s, $VR = 5.64$ s); the Wilcoxon on time moves from $p = 0.074$ to $p = 0.50$. The direction of the result depends heavily on B1 alone: of the three measures, only decision time survives B1’s removal as a consistent V-favoring signal, and even that signal becomes much weaker. This reinforces the case for revising B1 before further data collection.

Qualitative signal: prompt visibility. Three of nine participants independently noted that the reflective prompts were easy to overlook. In the post-session questionnaire (Appendix D.7) P4 said they did not read them at all, while P9 suggested making the

prompts bigger. This suggests that the reflective layer may not have failed because of its content, but because its delivery did not capture attention during the task.

4 Future Work

Several directions follow naturally from the present results.

From the experiment. The most direct next step is to complete the planned sample ($N=12$) and re-check the AB versus BA group means; if the asymmetric skill transfer reported in Section 3.2 persists at full sample size, the V versus VR comparison will need to be reframed accordingly. A second step is to revise scenario B1 so that its period ordering matches the rest of the set, then redraw a fresh random pair within the same temporal grain. A third direction is to test an *onboarding-versus-overlay* variant of VR, in which the reflective text appears once as a brief tutorial at the start of the session and then disappears, rather than persisting throughout.

From the usability evaluation. A participant suggested overlaying average listening patterns onto the visualizations as comparative reference points (Appendix C). While these suggestions were considered valuable, there was not enough time to add it to this prototype, but this idea should be implemented in the future.

From the prototype. Beyond the experiment, three engineering directions would extend the system’s reach. **Data integration:** extending support beyond Spotify to other streaming platforms (e.g., Apple Music, YouTube Music, or Tidal) would broaden the prototype’s audience. **Contextual data integration:** incorporating external signals such as calendar events or geolocation could let users correlate listening patterns with broader life events, moving the system closer to a fuller personal informatics experience. **Onboarding automation:** the current pipeline requires users to manually request their Spotify export, wait several days, and pre-process the raw files. Direct integration with the Spotify Web API would remove this friction. **Mobile delivery:** re-implementing the interface as a mobile application would better match the contexts in which music listening actually occurs.

5 Conclusion

This paper presented *A Life in Songs*, an interactive personal informatics prototype for reflecting on long-term Spotify listening history. The prototype was evaluated through a think-aloud usability evaluation and a within-subjects HCI experiment.

The usability evaluation drove substantial changes to the prototype, such as thematic dashboarding, simpler narrative language, and clearer interaction cues. It also confirmed that users with different levels of familiarity could navigate and extract meaning from the redesigned notebook. The HCI experiment did not support the hypothesis that reflective guidance would improve performance: under VR, participants were, on average, slower, slightly less accurate, and less confident than under V. However, the preliminary sample and counterbalancing effects mean that these findings should be interpreted cautiously.

Overall, the results suggest that reflective guidance may be more useful as short onboarding support than as a persistent overlay. Future work should test this distinction with a larger sample and a redesigned onboarding-based version of the prototype.

Acknowledgments

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A Observable Notebooks and GitHub Repository

- **Prototype (Observable Notebook):** [Link](#)
- **HCI Experiment (Observable Notebook):** [Link](#)
- **Data Preprocessing Pipeline and HCI Experiment Analysis (GitHub):** [Link](#)

B Design Alternatives Considered for the Focal Chart

Three alternative designs were sketched before settling on the polar clock. A linear *bar chart* (Figure 6) plots listening hours against hour-of-day along an x-axis; it is more readable for precise values but splits the 22h–04h late-night window into two disconnected segments at opposite ends of the axis, making period-to-period late-night comparison harder than necessary. A *calendar heatmap* (similar to the contributions-style chart already used elsewhere in the notebook) shows long-term patterns clearly, but collapses time-of-day into a single daily cell, so it cannot represent late-night listening at all. A *combined view*, inspired by date $\times$ time-of-day scatter plots paired with daily-totals bar charts seen in some symptom-tracking tools, was attractive in principle but added visual complexity without a clear gain over the polar clock for the comparison task at hand. The polar clock was chosen because it (i) maps the natural cyclicity of time-of-day to angle, and (ii) renders the late-night band as a single contiguous sector at the top rather than splitting it across the chart.

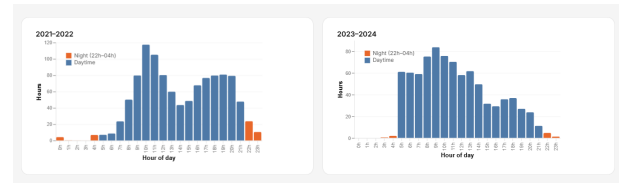


Figure 6: An alternative bar-chart design considered for the focal chart, comparing two periods (2021–2022 vs. 2023–2024). Readable for precise hour-by-hour values, but the 22h–04h late-night window is split between the two ends of the x-axis, which made period-to-period late-night comparison less direct than the polar clock layout eventually adopted.

C Usability evaluation

In this section, we gathered more detailed information of the usability evaluations, separated by participant. This includes each participant’s background in data visualization and programming, how their individual sessions went and their feedback during and after the session. We also included the main changes made to the prototype after each session.

Below is the list of tasks given to participants during the sessions.

Task 1 - Daily listening Intensity “Use the year range filter to show only 2020 and 2021. In which month of those years did you listen to the most music?”

Task 2- The Race Over Time “Switch the mode to Genres and press play. Which genre was in first place by the end of 2018?”

Task 3 - Your Listening Fingerprint “Set the year to 2022 and the month of January. Which genre dominated your listening that month?”

Task 4 - How You Listen “Select the time frame 2019–2021. In that period, did you listen more intentionally or through discovery?”

Task 5 - Music Across Borders “Which country did you listen to the most music in when abroad and how many hours does it show?”

Task 6 - Streaming Hours Explorer “Set the period slider to 2020–2022. At what time of day did you listen to music the most during those years?”

Now, we have the part regarding each session in particular.

Participant A (familiar with data visualization) found the notebook overwhelming, with no clear sense of progression between charts. They identified redundancy between some graphs and found some certain visualizations difficult to read, taking a lot of time to complete several tasks. This session led to the most significant structural changes. The notebook was reorganized into thematic **dashboards**, which improved visual coherence and allowed for the inclusion of additional relevant information, such as highlights (as seen in Figure 7). Consequently, some graphs were merged into a single linked view, and filters and interactions were made more explicit.

Participant B (low familiarity) relied heavily on visual hints and found several sections hard to follow. They spent a considerable amount of time on certain graphs, such as the stream graph

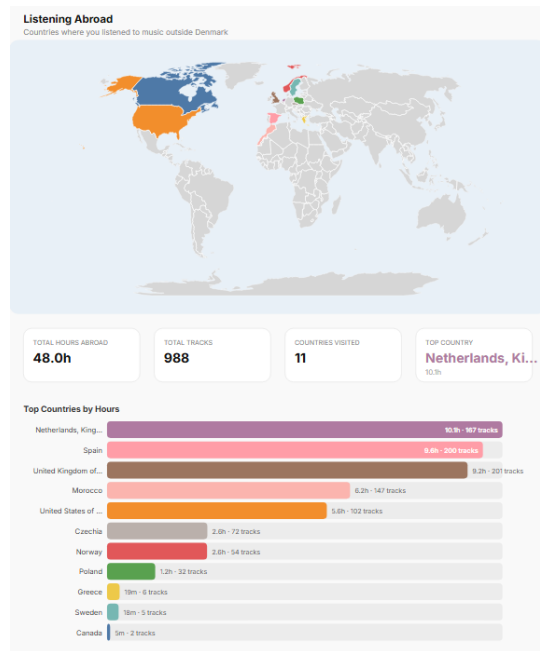


Figure 7: Example of a dashboard in the prototype

(shown in Figure 8) and the Discovery dashboard, without fully understanding them and responded most positively to the bar chart race. The stream graph was removed and replaced by a combined radial chart (Figure 9). The bar chart race was moved earlier in the notebook to build user confidence before introducing more complex views and all descriptions were rewritten in simpler and more narrative language, highlighting the key takeaways of each graph. In the discovery dashboard, more detailed explanations were added for each of the display metrics (Figure 10).

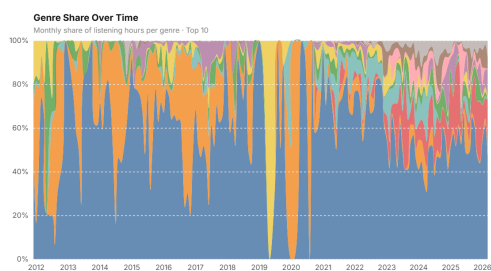


Figure 8: Stream Graph

Participant C (moderate familiarity) found the dashboard structure intuitive but struggled to track individual artists in the bump chart, due to over-plotting (as seen in Figure 11). Artist filtering was added to the bump chart to allow trajectory isolation and hover interaction was introduced to allow clarity even with a small number of artists (as seen in Figure 12). Transition sentences were introduced between sections to improve narrative flow.

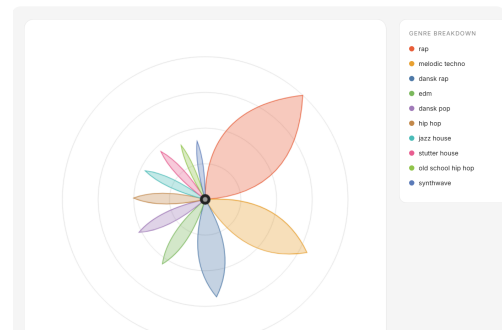


Figure 9: Radial Chart

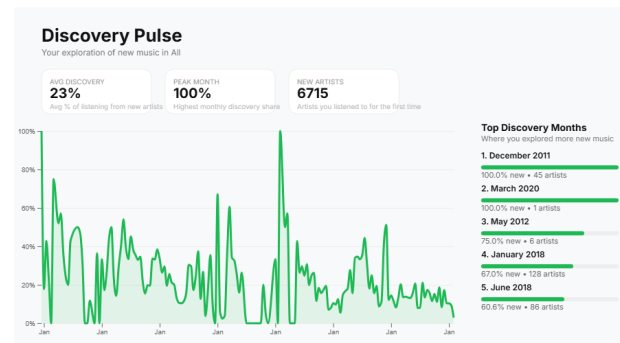


Figure 10: Discovery Dashboard updated

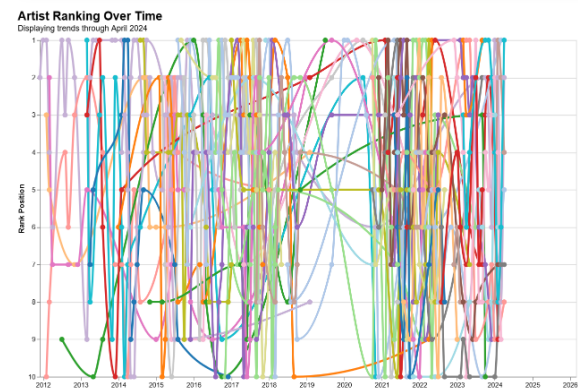


Figure 11: Artist Ranking chart before

Participant D (low familiarity) Had difficulty matching some graphs legends, which appeared far from the smaller screens and preferred using the bar chart race over the bump chart for artist tasks. All legends were repositioned to stay close to the charts at all screen sizes and some of the sections were rewritten for clarity.

Participant E (low familiarity) engaged slowly, reading every word before touching any control. Found the heatmap and bar chart race accessible, but was confused by the radial fingerprint chart and asked for more information regarding the scale of each petal.

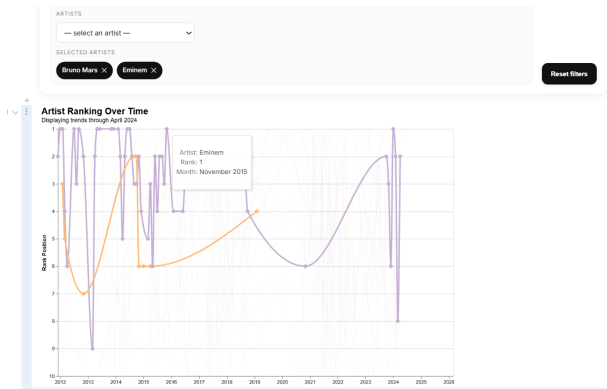


Figure 12: Artist Ranking chart after (with filters and hover interaction)

The fingerprint section description was rewritten to lead with a concrete explanation of what the shape encodes and hover interactions were added (as seen in Figure 13).

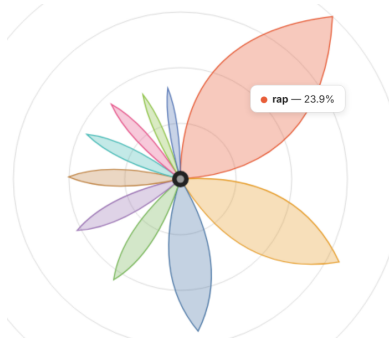


Figure 13: Radial Chart with hover interaction

Participant F (moderate familiarity) responded strongly to the heatmap and the geography section but felt the "How You Listen" and "Discovery vs. Comfort" sections were too abstract, meaning that the concepts of intentional versus discovery listening required more explanation before the chart made sense. Found some of the genre labels confusing without context. Introductory sentences were added to the behavior sections to ground the concepts before presenting the data, and a short tooltip was added to genre labels explaining each one on hover. It should be noted that the genre labels are retrieved directly via API and are not within the scope of this prototype to modify.

Participant G (high familiarity) approached the prototype fast and completed all tasks efficiently but raised some concerns regarding the lack of comparative context in several charts, giving the idea of adding to some of them the average listening patterns to comparison. Since this was the last usability evaluation and due to the lack of time we did not have the time to take this suggestion but it would be an improvement to this prototype.

D HCI Experiment

D.1 Session procedure

Each session followed the fixed sequence shown in Table 2, beginning with consent and demographics and closing with a post-questionnaire and open debrief.

Table 2: Session procedure overview

Time	Phase	What happens
~3 min	Consent and demographics	The participant completes a short questionnaire (age, Spotify usage per day, prior use of Wrapped or Last.fm, comfort with charts).
~3 min	Task explanation	The experimenter explains the task, shows one worked example, and answers procedural questions, but does not coach interpretation of the visualizations.
~2 min	Practice trial	The participant completes one practice scenario under whichever condition comes first. Practice data are not analyzed.
~8 min	Condition 1	The participant completes three scenarios under their first interface (V or VR). Decision time and answers are logged automatically, confidence is collected on screen after each scenario.
~8 min	Condition 2	The participant repeats the procedure under the second interface, using the parallel scenario set.
~5 min	Post-questionnaire and debrief	The participant rates the two interfaces on perceived helpfulness and gives open-ended feedback.

D.2 Choice of within-subjects design

A within-subjects design was chosen because each participant could act as their own baseline when comparing V and VR. This is useful in HCI experiments with small samples, as it reduces the influence of individual differences such as chart literacy, Spotify familiarity, or general speed in interpreting visualizations. It also requires fewer participants than a between-subjects design, while still giving each participant experience with both interface conditions [6, Ch. 5].

D.3 Data analysis plan

The three dependent measures are analyzed using paired tests, since each participant contributes data under both conditions:

- **Decision time:** per-participant mean times under V and VR are compared with a paired-samples *t*-test if the differences are approximately normal (Shapiro–Wilk), otherwise a Wilcoxon signed-rank test.
- **Error rate:** per-participant error counts (out of three) are compared with a Wilcoxon signed-rank test; for trial-level analysis we use McNemar’s test on matched correct/incorrect outcomes.
- **Confidence:** per-participant mean ratings are compared with a Wilcoxon signed-rank test, since Likert data are ordinal.

D.4 Participants

Table 3 summarizes the sample. All values are self-reported via the pre-session questionnaire (Appendix D.6). The Order column indicates whether the participant began with V (group AB) or VR (group BA).

Table 3: HCI experiment participants ($N=9$). W = prior Wrapped use, L = prior Last.fm use (Y/O/N = yes regularly / occasionally / no), Charts and Polar columns are 1–5 self-rated comfort.

ID	Age	Spotify (years)	Daily (h/day)	W	L	Charts (1–5)	Polar (1–5)	Order
P1	27	> 5y	1–2h	Y	O	4	4	AB
P2	26	3–5y	1–2h	Y	N	4	3	BA
P3	24	1–3y	2–4h	Y	N	4	4	AB
P4	27	> 5y	< 1h	Y	N	4	4	BA
P5	22	> 5y	1–2h	Y	Y	4	4	AB
P6	25	> 5y	< 1h	Y	N	4	4	AB
P7	23	3–5y	1–2h	Y	O	5	3	BA
P8	26	< 1y	1–2h	O	N	5	4	BA
P9	26	1–3y	1–2h	O	N	4	2	BA

D.5 Task scenarios

Table 4: Practice and experimental scenarios. “Truth” indicates which period contained the higher share of late-night listening; “Order” indicates whether Period 1 is chronologically earlier (E) or later (L) than Period 2; “Acc.” is per-scenario accuracy across the 9 participants run to date.

ID	Period 1	Period 2	Truth	Order	Acc.
Practice	Jan 2019	Dec 2019	P1	E	–
A1	Jan–Apr 2022	May–Aug 2022	P1	E	100%
A2	2021	2023	P1	E	77.8%
A3	Jan–Jun 2020	Jul–Dec 2020	P2	E	100%
B1	Sep–Dec 2023	Jan–Apr 2023	P2	L	66.7%
B2	2022	2024	P1	E	100%
B3	Jan–Jun 2021	Jul–Dec 2021	P1	E	88.9%

Figures 14 and 15 show examples of the two interface conditions. The V condition presents only the two polar clocks, while the VR condition adds reflective prompts and late-night summaries to guide interpretation.

Although the periods themselves were chosen randomly, the accuracy per-scenario reveals that Set B was systematically harder than Set A: Set A averaged 92.6% accuracy across its three scenarios, Set B averaged 85.2%, and within Set B the spread is wider (67–100%) than within Set A (78–100%). The clearest case is B1, with only 67% accuracy. Inspection of the list of scenarios reveals a structural reason: B1 is the only scenario in which Period 1 is chronologically *later* than Period 2. The other six pairs all present the earlier period on the left, which appears to have created an implicit “left = earlier” expectation by the time participants reached B1. Combined with B1’s ground-truth answer being P2 (the chronologically earlier of

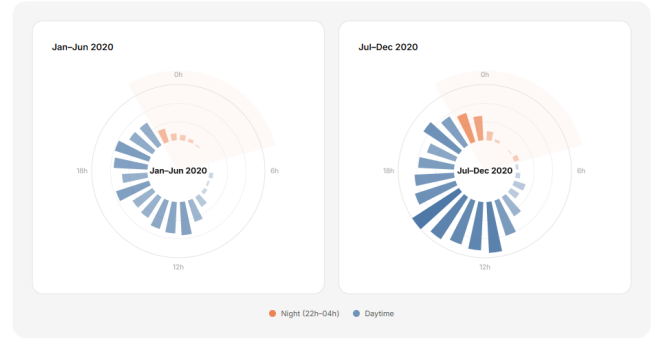


Figure 14: V condition (scenario A3): two polar clocks side by side, no additional text.

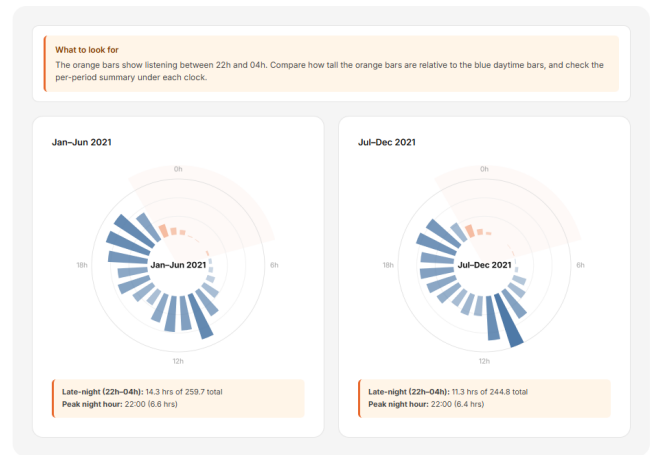


Figure 15: VR condition (scenario B3): the same chart layout augmented with a “What to look for” callout above and a per-period late-night summary beneath each clock.

the two), the scenario likely measured adherence to that expectation as much as late-night identification.

Figure 16 illustrates this issue: B1 reverses the chronological order used in all other scenarios.

This is a design flaw rather than an inherent difficulty difference: the random selection of periods produced one pair (B1) whose layout violated an expectation that the other six pairs had implicitly trained. B1 should be revised before further data collection so that all scenarios are chronologically consistent in their period ordering, after which a fresh randomization can be drawn. The current results in Section 3.2 are reported with this caveat.

D.6 Pre-session questionnaire

The following items were administered before training, on paper:

- (1) Age (open numeric).
- (2) Length of Spotify use (4 options: <1y, 1–3y, 3–5y, >5y).
- (3) Average daily Spotify listening (4 options: < 1h, 1–2h, 2–4h, >4h).
- (4) Prior use of Spotify Wrapped (3 options: Yes, every year, Yes, occasionally, No, never).

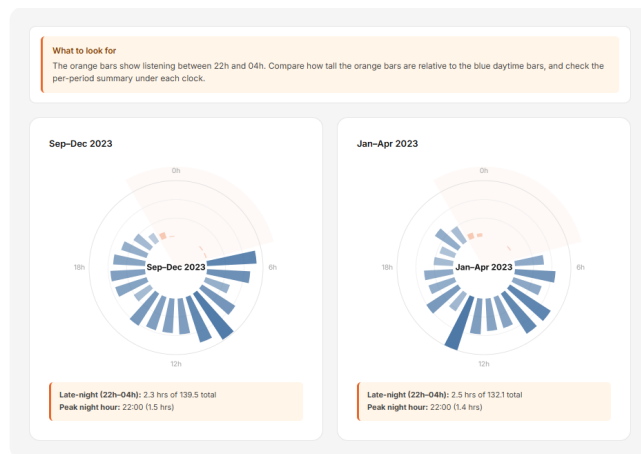


Figure 16: Scenario B1, the only scenario in which Period 1 (Sep–Dec 2023) is chronologically *later* than Period 2 (Jan–Apr 2023). The other six scenarios place the earlier period on the left, and B1’s reversal coincides with its anomalously low accuracy of 67%.

- (5) Prior use of Last.fm or comparable analytics tools (3 options).
- (6) Self-rated comfort with charts and data visualizations (1–5).
- (7) Self-rated comfort with circular/polar charts (1–5).

D.7 Post-session questionnaire

After both conditions, participants answered:

- (1) Which version felt easier to use? (V1 / V2 / no difference)
- (2) Which version did you trust more? (V1 / V2 / no difference)
- (3) Helpfulness of the reflective prompts (1–5).
- (4) Distractedness of the reflective prompts (1–5).
- (5) Open feedback on confusing or frustrating elements.
- (6) Free additional comments.

E AI Declaration

We used AI assistants — specifically Claude (Anthropic), ChatGPT (OpenAI), GitHub Copilot — at several stages of this project. Their use is summarized below for transparency.

Ideation. We utilized AI to help facilitate the ideation of our final project. By providing the course details and context, personal interests, and feasibility, we sparred with AI to decide on the topic and direction of our project. An example of such an ideation prompt would be:

Given the course context, could we do some ideation for an interesting final project. I’m not sure about the best way to do this ideation, but I’d think that we’d start broad, you’d ask me some questions, and we’ll specify and hone in on a direction. How does that sound?

With the above prompt, the AI (Claude in this case), proceeded to ask follow-up questions about personal interests, direction, and project priorities.

Code. AI was used to assist in the Python data-processing pipeline described in Section 1.2. Specifically, it helped clean the data and

enrich them with artist and genre metadata retrieved with the Spotify API. Additionally, it drafted boilerplate code for the Observable Notebook visualizations for us to iterate upon based on our usability testing and HCI experiment.

Statistical analysis. The Jupyter Notebook used to compute the descriptive statistics, the Wilcoxon signed-rank tests, the normality checks, and the sensitivity analysis (excluding scenario B1) reported in Section 3.2 was co-developed with AI assistance. The interpretation of the results, the framing of the caveats, and the conclusions drawn are the authors’ own.

Writing. AI was used to improve clarity, fix grammar, and suggest rewording of this report. The research questions, the study design, the choice of methods, the interpretation of qualitative and quantitative findings, and all discussion text were authored by the team. AI output was edited rather than accepted verbatim; the authors take full responsibility for the final wording.

Pros of the use of AI. The use of generative AI significantly sped up our implementation part of our project, i.e., coding the Python scripts and parts of the Observable Notebook. By delegating most of these tasks to the AI, it was easier to execute on the group’s vision for the prototype and system. It freed up time to think more creatively about the solution and quickly prototype new ideas for visualizations, and discard them if necessary, without extensive time commitments.

Cons of the use of AI. Debugging proved to be moderately more difficult in the AI-assisted parts of the codebase. Since some of the generated code was not written manually by the group, it occasionally required extra effort to understand the logic, trace errors, and ensure that the implementation aligned with our intended design.

F Author Contributions

Leonor led the usability evaluation, including participant sessions and analysis of usability findings. Anna designed and conducted the HCI experiment, including the experimental setup, data collection, and analysis. Adam led the data pipeline and supported the selection and preparation of the visualizations. All group members contributed equally to project ideation, report writing, and prototype implementation.